House Price Prediction Analysis

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Project Objective

The objective of this project was to build machine learning models for predicting house prices and to identify the housing features that have the greatest impact on selling price.

Dataset Overview

The study utilized the Housing Prices Dataset sourced from Kaggle.

- 545 observations.
- 13 housing features.
- Includes both numerical and categorical variables.
- Target variable: Price.

Data Preparation

The dataset was prepared before model training through a series of preprocessing steps.

- No missing values or duplicate records were found.
- Categorical features were converted using one-hot encoding.
- Relevant features were retained for model building.

Machine Learning Models

Two supervised regression models were trained:

- Linear Regression.
- Random Forest Regressor.

Model Performance

The models were evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the R-squared (R^2) coefficient.

Model	MAE	RMSE	R^2
Linear Regression	970,043	1,324,507	0.653
Random Forest	1,021,546	1,400,566	0.612

For this dataset, the Linear Regression model achieved lower prediction errors and a higher R^2 score than the Random Forest model, making it the better-performing approach.

Visual Analysis

Three key visualizations were generated to interpret the data and model outcomes:

- Histogram of house prices.
- Correlation heatmap.
- Actual vs Predicted scatter plot.

Key Findings

- **Primary Drivers:** Property area and the number of bathrooms demonstrate the strongest positive correlation with house price.
- **Influential Amenities:** Air conditioning and the number of stories showed positive relationships with house price.
- **Model Comparison:** Linear Regression outperformed the Random Forest Regressor on this specific dataset, achieving a higher R^2 value and lower error metrics.

Business Recommendation

Based on the analysis, real estate businesses should focus more on property area, number of bathrooms, and important amenities such as air conditioning when estimating house prices. These factors showed stronger relationships with selling price and can improve pricing decisions.

Conclusion

This project demonstrated that machine learning can be used to estimate house prices with reasonable accuracy. Among the two models tested, Linear Regression performed better for this dataset. The analysis also showed that property size and key amenities play an important role in determining house prices.